Assignment (17%)

1. A dataset of dissolved oxygen (DO, mg/L) measurements from a river station is given:

DO values	6.2	5.8	6.5	7.0	5.5	6.8	6.0	5.9

Tasks:

a. Compute the **mean, median, and mode** (if any).
b. Calculate the **variance and standard deviation**.
c. Interpret what the results indicate about water quality stability.

Answer:

Dataset:

6.2, 5.8, 6.5, 7.0, 5.5, 6.8, 6.0, 5.9

Number of observations (n) = 8

a. Mean, Median, and Mode

1. Mean (average)

\Mean = $\frac{\text{DO values}}{\mathbf{n}} = \frac{\mathbf{6.2+5.8+6.5+7.0+5.5+6.8+6.0+5.9}}{8} = \frac{49.7}{8}$= 6.2125 mg/L

2. Median (middle value)

Steps:

First, sort the data in ascending order:

5.5, 5.8, 5.9, 6.0**,** 6.2**, 6.5, 6.8, 7.0**

Since n = 8 (even), the median is the average of the 4th and 5th values:

Media = $\frac{\mathbf{6.0+\ 6.2}}{\mathbf{2}}$ = 6.1

3. Mode (most frequent value)

Each value appears exactly once.

Mode: None (no repeated values).

b. Variance and Standard Deviation

We use the **sample variance** (since this is a sample of measurements, not the entire population).

Step 1 – Find deviations from the mean (6.2125) and square them:

DO (x)	x − mean	(x − mean)²
6.2	−0.0125	0.000156
5.8	−0.4125	0.170156
6.5	0.2875	0.082656
7.0	0.7875	0.620156
5.5	−0.7125	0.507656
6.8	0.5875	0.345156
6.0	−0.2125	0.045156
5.9	−0.3125	0.097656

Step 2 – Sum of squared deviations:

$\sum(x-\bar{x})^2$=0.000156+0.170156+0.082656+0.620156+0.507656+0.345156+0.045156+0.097656= 1.86875=1.86875

Step 3 – Sample variance (s^2):

$$\mathbf{s^2 = \frac{1.86875}{8} = 0.26696\ (mg/L)^2}$$

Step 4 – Sample standard deviation (s):

$$s=\sqrt{0.26696} \approx 0.5167 \text{ mg/L}$$

c. Interpretation of Water Quality Stability

- **Mean DO = 6.21 mg/L** – This is well above the typical minimum acceptable level for healthy aquatic life (usually $\geq$ 5 mg/L for many species). So, on average, the water is adequately oxygenated.
- **Standard deviation = 0.52 mg/L** – This is relatively small compared to the mean (about 8.3% of the mean). It indicates that most measurements are clustered closely around 6.2 mg/L.
- **Range** = 7.0 − 5.5 = 1.5 mg/L – This is a narrow range, showing no extreme fluctuations.

Overall interpretation:

The DO levels are **stable** and **consistent** over these measurements.

There is low variability, meaning the river station does not experience sudden drops or spikes in dissolved oxygen. This suggests good short-term water quality stability, with no signs of pollution pulses or algal bloom crashes that would cause large swings in DO. However, since this is a small sample over an unspecified time period, **long-term trends would require more frequent sampling.**

2. PM-2.5 concentration (µg/m³) in a city over 10 days is recorded as:

12, 15, 18, 20, 22, 25, 28, 30, 35, 80

Tasks:

a. Describe the distribution of the dataset.
b. Identify any **outliers** and justify your answer.
c. Explain how outliers can affect environmental decision-making.
d. Suggest one statistical method to handle outliers.

Answer:

Dataset:

12, 15, 18, 20, 22, 25, 28, 30, 35, 80

Number of observations (n) = 10

a. Describe the distribution of the dataset

1. Shape – Skewness

- Most values (9 out of 10) are clustered between **12 and 35** µg/m³.
- One value (**80**) is far to the right of the rest.
- This creates a **right-skewed (positively skewed)** distribution, where the long tail extends toward higher values.

2. Center

- **Mean** $= \frac{\mathbf{12+15+18+20+22+25+28+30+35+80}}{\mathbf{10}}$ = **28.5** µg/m³
- **Median** (average of 5th and 6th sorted values) $= \frac{\mathbf{22+25}}{\mathbf{2}}$ = **23.5** µg/m³
- The mean (28.5) is larger than the median (23.5), which is a classic sign of right-skewness.

3. Spread

- **Range** = 80 − 12 = **68** µg/m³ (very wide, driven by the high value).
- Most values lie within 12–35, so the bulk of the data is relatively tight, but the single 80 expands the range dramatically.

Overall description:

The dataset is **asymmetric with a strong positive skew**, having one unusually high value that pulls the mean upward. The typical central value is better represented by the median (23.5) than the mean.

b. Identify any outliers and justify your answer

We use the **Interquartile Range (IQR) method**, which is robust to skewness.

Step-1 – Find Q1 and Q3

Sorted data: 12, 15, 18, 20, 22, 25, 28, 30, 35, 80

- **Q1** (25^{th} percentile) = median of first 5 values (12,15,18,20,22) = **18**
- **Q3** (75^{th} percentile) = median of last 5 values (25,28,30,35,80) = **30**

Step 2 – Compute IQR

IQR = Q3 − Q1 = 30 − 18 = **12**

Step 3 – Determine outlier boundaries

- Lower fence = Q1 − 1.5×IQR = 18 − 18 = **0**
- Upper fence = Q3 + 1.5×IQR = 30 + 18 = **48**

Step 4 – Identify outliers

Any value < 0 or > 48 is an outlier.

- **80 is > 48**, so **80 μg/m³ is an outlier**.

Justification:

The value 80 is more than 1.5×IQR above the third quartile, meaning it is unusually far from the main cluster of data. It is also more than 3 standard deviations from the mean (mean=28.5, sd≈19.4 → 28.5+3×19.4=86.7, but IQR is preferred for skewness).

c. How outliers can affect environmental decision-making

Area	**Effect of the outlier (80 μg/m³)**
Regulatory compliance	If the air quality standard is, say, 24-hour average of 50 μg/m³, the mean (28.5) is safe, but the outlier might trigger a false alarm if a single day is compared to a daily limit.

Public health warnings	A single extreme spike might cause unnecessary panic or, conversely, if averaged with low values, it may hide the fact that one day was very unhealthy.
Policy planning	Long-term averages used for pollution control strategies will be overestimated (28.5 vs. a more typical 23.5), potentially leading to misallocation of resources (e.g., overly strict or lenient measures).
Trend analysis	An outlier can distort year-over-year comparisons, making it look like pollution increased when it was just one anomalous event (e.g., a fire or dust storm).
Risk assessment	Chronic exposure (long-term) is more harmful than acute spikes; focusing on the outlier might distract from the consistently moderate levels that actually pose greater cumulative risk.

d. Suggest one statistical method to handle outliers

Method: Winsorization (or trimming)

- **Winsorization** replaces the extreme outlier with the next highest value that is not an outlier. For this dataset, you would replace 80 with the nearest non-outlier (35 or the upper fence value 48).
- **Example** (cap at 48): New dataset: 12,15,18,20,22,25,28,30,35,48 → mean becomes 25.3, which better reflects the bulk of the data.
- **Alternative (Trimming):** Remove the outlier entirely (leaving 9 values) and use the median or mean of the remaining data.

Why choose Winsorization?

- It preserves the sample size (n=10) for statistical tests.
- It reduces the influence of the outlier without discarding it completely, acknowledging that a high value did occur but adjusting it to a more plausible maximum for analysis.

Other valid methods (you could mention):

- Log-transformation (to reduce skewness)
- Use robust statistics (median instead of mean, IQR instead of standard deviation)
- Investigate the source (e.g., was it a sensor error or a real event?) before deciding

3. You are developing a machine learning model to predict soil pH using environmental variables such as rainfall, temperature, and soil nutrients.

Tasks:

a. Define the **problem type** (regression/classification) and justify.
b. List at least **4 possible input features**.
c. Explain the role of **data preprocessing** in improving model performance.
d. Discuss two suitable algorithms for this task and justify your choice.

Answer:

Here is the step-by-step solution for developing a machine learning model to predict soil pH.

a. Define the problem type (regression/classification) and justify

Problem type: Regression

Justification:

- **Soil pH is a continuous numerical variable** (typically ranging from 0 to 14, with values like 5.3, 6.8, 7.2, etc.).
- The goal is to **predict a specific numeric value** rather than assigning a categorical label.
- If the task were classification, the output would be discrete categories such as "acidic" (pH < 6.5), "neutral" (6.5–7.5), or "alkaline" (pH > 7.5). However, since the prompt asks to predict pH itself (not a category), regression is the correct framing.
- Evaluation metrics for regression (e.g., RMSE, MAE, R^2) are appropriate here.

b. List at least 4 possible input features

Feature	Description	Why it matters for pH
Rainfall (mm/year)	Annual or seasonal precipitation amount	Rain leaches basic cations (Ca^{2+}, Mg^{2+}) and can acidify soil over time; also affects organic matter decomposition.

Temperature (°C)	Mean annual or growing-season temperature	Higher temperatures accelerate microbial activity and organic matter breakdown, releasing organic acids that lower pH.
Soil Organic Matter (%)	Percentage of organic carbon in soil	Decomposing organic matter produces humic and fulvic acids, which naturally acidify the soil.
Nitrogen (N) concentration (mg/kg)	Available nitrate or ammonium content	Nitrogen fertilizers (especially ammonium-based) undergo nitrification, releasing H^+ ions and lowering pH.
Parent Material Type	Categorical (e.g., granite, limestone, basalt)	Parent rock determines the buffering capacity; limestone-rich soils are alkaline, while granite-rich soils tend to be acidic.
Elevation (m)	Height above sea level	Influences temperature, rainfall patterns, and drainage, which indirectly affect pH.
Cation Exchange Capacity (CEC)	Soil's ability to retain cations	High CEC usually indicates higher buffering capacity, resisting pH changes.

c. Explain the role of data preprocessing in improving model performance

Data preprocessing is **critical** because real-world environmental data is messy. Without it, model performance degrades significantly. Key preprocessing steps and their roles:

Preprocessing Step	**Role in Improving Performance**
Handling Missing Values	Soil data often has gaps due to sensor failures or lab errors. Imputation (e.g., mean, median, or KNN-based) prevents the model from discarding entire samples, preserving dataset size and statistical power.
Outlier Detection/Capping	Extreme values (e.g., a sensor error reading pH = 14.5) can skew the model's learning. Capping or removing outliers prevents the model from "overfitting" to rare, unrealistic events.

Preprocessing Step	Role in Improving Performance
Feature Scaling (Normalization/Standardization)	Features have different units (rainfall in mm, temperature in °C, nutrients in mg/kg). Algorithms like SVM or neural networks are distance-based; scaling ensures no single feature dominates solely due to its magnitude.
Encoding Categorical Variables	Features like "parent material" or "soil type" need to be converted to numeric (one-hot or label encoding) so the model can process them.
Feature Engineering	Creating new features (e.g., "rainfall × temperature" interaction term) can capture complex environmental processes that individual features miss, improving predictive power.
Data Splitting (Train/Validation/Test)	Ensures the model is evaluated on unseen data, preventing overfitting and providing a realistic estimate of real-world performance.
Addressing Class/Value Imbalance	If the dataset has few alkaline soil samples, the model may ignore them. Techniques like SMOTE (for regression variants) or weighted loss functions ensure balanced learning across the pH spectrum.

d. Discuss two suitable algorithms for this task and justify your choice

Algorithm 1: Random Forest Regressor

Why it is suitable:

- **Handles non-linearity well:** Soil pH is influenced by complex interactions (e.g., rainfall × temperature × organic matter). Random Forest captures these without requiring manual specification.

- **Robust to outliers and noise:** Environmental data is noisy; Random Forest uses bootstrapping and averaging, making it resistant to overfitting.
- **Feature importance:** It provides built-in ranking of which variables (e.g., rainfall vs. nitrogen) most affect pH, giving agronomists actionable insights.
- **No scaling required:** Being tree-based, it is invariant to feature magnitudes, so you can skip normalization.
- **Works with mixed data types:** Can handle both continuous (rainfall, temp) and categorical (soil type, parent material) features natively.

Best for: Medium-sized datasets (hundreds to thousands of samples) where interpretability and robustness are priorities.

Algorithm 2: Gradient Boosting (e.g., XGBoost or LightGBM)

Why it is suitable:

- **State-of-the-art predictive accuracy:** Gradient boosting sequentially corrects errors from previous trees, often yielding the highest R^2 scores on tabular environmental data.
- **Handles complex interactions:** Like Random Forest, it captures non-linearities and feature interactions automatically.
- **Regularization built-in:** XGBoost has L1/L2 regularization parameters that prevent overfitting, crucial when you have many correlated soil nutrient features.
- **Efficient with missing values:** XGBoost can learn the best direction to handle missing data during training, reducing the need for heavy imputation.
- **Scalability:** LightGBM, in particular, is extremely fast and memory-efficient, making it suitable for large-scale national soil surveys.

Best for: Larger datasets (thousands+ samples) where maximum predictive performance is required, even at the cost of slightly less interpretability than Random Forest.

Alternative honorable mention: Support Vector Regression (SVR) with RBF kernel

- Suitable if the dataset is small (< 500 samples) and the relationship is highly non-linear. However, it requires careful scaling and tuning, and is less interpretable than tree-based models.

Final Recommendation for Your Project:

Start with **Random Forest** as a baseline because it is easy to tune, interpretable, and works well out-of-the-box. Then, if you need better accuracy and have sufficient data, move to **XGBoost** and fine-tune hyperparameters (learning rate, tree depth, regularization) to push performance further. Always validate using cross-validation on a held-out test set.

4. A simple linear regression model is used to predict PM2.5 concentration based on temperature (°C):

Temperature	**PM2.5**
20	30
25	28
30	25
35	20

Tasks:

a. Plot and describe the relationship between temperature and PM2.5.
b. Determine whether the relationship is positive or negative.
c. Fit a conceptual linear regression model:
   (y = mx + c)
d. Predict PM2.5 when temperature = 28°C (show working or estimation logic).

Answer:

Dataset:

Temperature (x): 20, 25, 30, 35

PM2.5 (y): 30, 28, 25, 20

Number of observations (n) = 4

a. Plot and describe the relationship between temperature and PM2.5

Scatter Plot Description:

If we plot Temperature on the x-axis (20 to 35) and PM2.5 on the y-axis (20 to 30), the points would be:

- (20, 30)
- (25, 28)
- (30, 25)
- (35, 20)

Visual pattern: As you move from left to right along the x-axis, the points consistently go downward. They form a roughly straight line that slopes from the top-left (high PM2.5 at low temperature) to the bottom-right (low PM2.5 at high temperature).

Relationship description: There is a **strong, linear, inverse (negative) relationship**. When temperature increases, PM2.5 concentration tends to decrease in a nearly perfectly straight-line fashion.

b. Determine whether the relationship is positive or negative

The relationship is NEGATIVE.

Justification:

- As Temperature (x) increases, PM2.5 (y) consistently decreases.
- The slope: $\frac{\textbf{change in y}}{\textbf{change in x}}$ is negative. For example:
  - From 20°C to 25°C ($\Delta x = +5$), PM2.5 drops from 30 to 28 ($\Delta y = -2$)
  - From 30°C to 35°C ($\Delta x = +5$), PM2.5 drops from 25 to 20 ($\Delta y = -5$)
- In all cases, the variables move in opposite directions.

c. Fit a conceptual linear regression model: y=mx+c$y=mx+c$

We need to find the slope (m) and y-intercept (c) using the **least squares method**.

Step 1 – Calculate necessary sums:

x (Temp)	y (PM2.5)	x·y	x^2
20	30	600	400
25	28	700	625
30	25	750	900
35	20	700	1225
Σx = 110	Σy = 103	Σxy = 2750	$\Sigma x^2 = 3150$

Step 2 – Calculate means:

$$\bar{x}=\frac{110}{4}=27.5,$$

$$\bar{y}=\frac{103}{4}=25.75$$

Step 3 – Calculate slope (m):

$$m=\frac{n(\Sigma xy)-(\Sigma x)(\Sigma y)}{n(\Sigma x)^2-(\Sigma x)^2}$$

$$m=\frac{4(2750)-(110)(103}{4(3150)-(110)^2}$$

$$=\frac{11000-11330}{12600-12100}=-0.66m$$

Step 4 – Calculate y-intercept (c):since y=mx+c

$$c=\bar{y}-m\bar{x}=25.75-(-0.66)(27.5)$$

$$c=25.75+18.15=43.90$$

Final fitted regression model:

$y = -0.66x + 43.90$

(Where y = predicted PM2.5, x = temperature in °C)

d. Predict PM2.5 when temperature = 28°C

Using the fitted model:

$$\mathbf{y = -0.66(28) + 43.90}$$

$$\mathbf{y = -18.48 + 43.90 = 25.42}$$

Prediction:

$$\mathbf{PM_{2.5} \approx 25.4\ \mu g/m^3 \text{ when temperature is } 28^{\circ}C}\ \boldsymbol{C}$$

Working/Estimation logic check:

- At 25°C, actual PM2.5 = 28
- At 30°C, actual PM2.5 = 25
- 28°C is 3/5 of the way from 25°C to 30°C (60% of the interval).
- The drop from 28 to 25 over 5°C is −3 units. 60% of −3 = −1.8.
- So estimated PM2.5 = 28 − 1.8 = 26.2 (using linear interpolation between actual points).
- Our regression gives 25.4, which is slightly lower because the regression line is "averaged" over all points and accounts for the steeper drop at higher temperatures. Both are reasonable, but the regression estimate is statistically optimal.

5. Explain how machine learning can be used in **environmental pollution monitoring and prediction**.

Answer:

Machine learning (ML) has revolutionized environmental monitoring by enabling the analysis of massive, complex, and non-linear datasets that are not handled by traditional statistical models .

Here is a comprehensive explanation of how ML is used in environmental pollution monitoring and prediction, broken down by applications, specific algorithms, and real-world impact.

1. Key Applications of ML in Pollution Monitoring

A. Spatiotemporal Prediction (Forecasting)

- **What it does:** Predicts pollution levels (PM2.5, NO_2, O_3) hours, days, or even weeks in advance.
- **How:** ML models learn from historical weather data (wind speed, temperature, humidity), traffic patterns, and past pollutant concentrations to forecast future spikes.
- **Example:** Predicting tomorrow's PM2.5 levels in a city to issue early health advisories for vulnerable populations (elderly, asthmatics).

B. Source Apportionment (Identifying Pollution Sources)

- **What it does:** Determines whether pollution comes from traffic, industrial factories, agricultural burning, or natural dust.
- **How:** ML algorithms analyze the "chemical fingerprint" (ratios of trace gases and particulate matter components) to classify the dominant source in real-time.
- **Example:** Distinguishing between a local traffic jam spike and a long-range dust storm traveling from a desert.

C. Anomaly Detection (Event Alerts)

- **What it does:** Identifies sudden, unexpected pollution events or sensor malfunctions.
- **How:** Unsupervised ML (like Autoencoders) learns the "normal" daily cycle of pollution. When readings deviate significantly from this learned pattern, it flags an alert.
- **Example:** Detecting an illegal toxic gas release from a nearby factory within minutes, rather than hours.

D. Filling Missing Data (Imputation)

- **What it does:** Reconstructs missing or corrupted data from faulty monitoring stations.
- **How:** ML uses spatial correlations (data from nearby stations) and temporal correlations (data from previous days) to intelligently guess the missing values, ensuring continuous datasets for regulators.

E. Satellite and Remote Sensing Calibration

- **What it does:** Converts raw satellite aerosol optical depth (AOD) readings into ground-level PM2.5 concentrations.
- **How:** ML bridges the gap between what a satellite sees in the atmosphere and what people actually breathe at ground level, which is crucial for regions with no ground monitors.